PHARMACOLOGY

Drugs in Bowel Diseases

SGL 2
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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What distinguishes ulcerative colitis from Crohn's disease in terms of inflammation pattern and GI tract involvement?
2. Why does sulfasalazine cause more systemic side effects than mesalamine, olsalazine, or balsalazide?
3. Which biologic class carries a notable risk of hepatitis B reactivation and tuberculosis reactivation?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Ulcerative colitis vs Crohn's disease

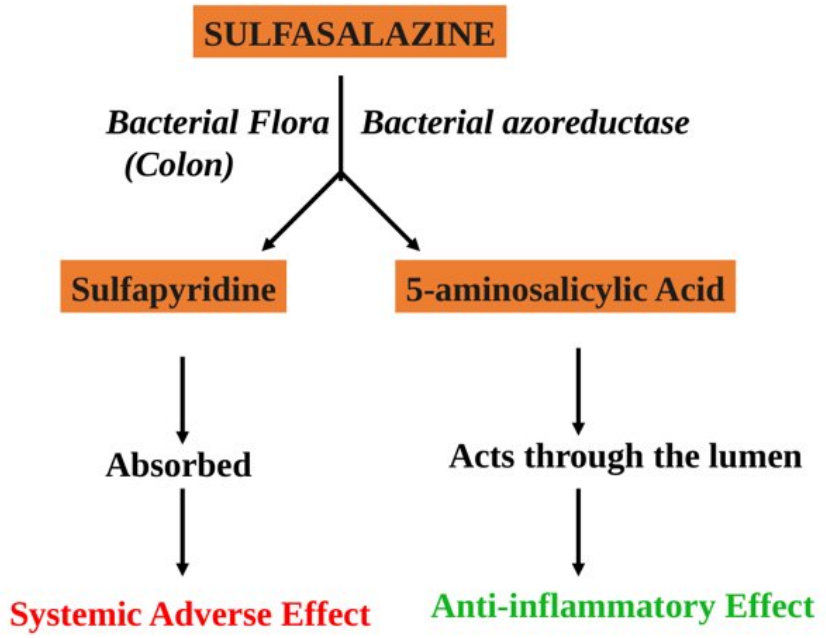
Feature	Ulcerative colitis	Crohn's disease
Location	Colon and rectum	Any part of the GI tract
Rectum	Universally involved	Involved in only 10% of cases
Depth of injury	Mucosal and submucosal	Transmural — may lead to strictures or fistulae, incl. perianal
Pattern	Continuous inflammation	Skip lesions (patchy)
Onset	Acute	Insidious

~10% of cases don't fit either pattern and are deemed indeterminate colitis. Clinical features shared by both: bloody diarrhoea, mucus in stools, urgency, tenesmus, malaise, weight loss, fever.

2 Drug classes used in IBD

Class	Drugs
Aminosalicylates	Sulphasalazine, mesalamine, olsalazine, balsalazide
Glucocorticoids	Prednisolone, methylprednisolone, hydrocortisone, budesonide
Immunomodulators	Azathioprine, 6-mercaptopurine, methotrexate, cyclosporine
Biological response modifiers	Infliximab, adalimumab, certolizumab, vedolizumab, ustekinumab
JAK inhibitors	Tofacitinib (UC), upadacitinib (UC and Crohn's)
Antibiotics	Metronidazole, ciprofloxacin, clarithromycin
Others	Probiotics, S1P receptor modulators

3 Aminosalicylates (5-ASA)



Sulfasalazine is split by colonic bacteria into sulfapyridine (absorbed → systemic ADRs) and 5-ASA (acts through the lumen → anti-inflammatory effect).

Sulphasalazine	Prodrug of sulfapyridine + 5-ASA. 5-ASA inhibits inflammatory mediators (COX/LOX, neutrophil chemotaxis, IL-1/TNF-α) locally. Sulfapyridine is absorbed and causes nausea, vomiting, headache, bone marrow suppression, rarely oligospermia; allergic reactions (rash, fever, hepatitis, pancreatitis, pneumonitis).
Mesalamine, olsalazine, balsalazide	Special formulations/prodrugs that deliver 5-ASA to the colon with fewer side effects than sulphasalazine. Olsalazine commonly causes diarrhoea. All may cause headache and skin rashes (incl. Stevens-Johnson syndrome). Mainly effective for mild-to-moderate UC.
Indications	Induce remission in mild UC/CD (higher doses); maintain remission in UC (less effective for maintenance in CD); long-term use in extensive UC reduces colorectal cancer risk by ~75%.

4 Other IBD drug classes

Glucocorticoids	Short-term induction of remission in moderate-severe IBD, not for maintenance. Prednisolone (oral), methylprednisolone (oral/parenteral), hydrocortisone (enema/suppository), budesonide (oral, high first-pass → fewer systemic effects). MOA: bind cytosolic receptors, suppress pro-inflammatory gene transcription (IL-2, TNF-α, IFN-γ). ADRs: HPA-axis suppression, osteoporosis, peptic ulcer, infections, hyperglycaemia, Cushingoid features.
Thiopurines (azathioprine, 6-MP)	For severe/steroid-dependent IBD. MOA: converted to 6-thioguanine nucleotides, block purine synthesis and DNA/RNA synthesis in lymphocytes. Delayed onset 8–12 weeks. Metabolized by TPMT — polymorphisms raise toxicity. ADRs: myelosuppression, hepatotoxicity, pancreatitis, increased lymphoma risk.
Methotrexate	Oral/IM/SC weekly. Inhibits dihydrofolate reductase → increases adenosine, decreases inflammation. More effective in Crohn's than UC. ADRs: bone marrow suppression, hepatotoxicity, pulmonary fibrosis.
Anti-TNF (infliximab, adalimumab, certolizumab, golimumab)	For severe Crohn's/refractory UC. MOA: bind TNF-α, block TNFR interaction. Costly, increased infection risk. ADRs: infusion reactions, infections (TB, hepatitis B reactivation), demyelination, lymphoma.
Vedolizumab (anti-integrin)	α4β7 integrin antibody. MOA: blocks lymphocyte trafficking. Safer systemic profile, rare PML risk — much lower than natalizumab.
Ustekinumab (anti-IL-12/23)	Binds p40 subunit of IL-12/IL-23, inhibits Th1/Th17 pathways. For moderate-severe Crohn's and UC. ADRs: infections, rare malignancy.

JAK inhibitors	Tofacitinib (UC), upadacitinib (UC and Crohn's) — oral. MOA: inhibit JAK1/3, block STAT signalling and cytokine-driven inflammation. ADRs: infections (herpes zoster), hyperlipidaemia, cytopenias, rarely thrombosis.
S1P modulators (ozanimod & class)	Ozanimod, fingolimod, siponimod, ponesimod, etrasimod. MOA: reduce lymphocyte egress from lymph nodes (S1P1/S1P5 for ozanimod) → less circulation/inflammation. Used in MS and (ozanimod) moderate-severe UC. Watch: bradycardia, liver function, BP, infections, macular oedema.

5 Probiotics

Live microorganisms (*Lactobacillus*, *Streptococcus thermophilus*, *Bifidobacterium*, *Saccharomyces boulardii*) that confer benefit by improving/restoring gut flora. Formulations: capsules/tablets, sachets, yoghurt/fermented milk, fortified foods, synbiotics (probiotic + prebiotic, e.g. inulin).

Therapeutic use	Effect
Antibiotic-associated diarrhoea	Restores gut flora, reduces C. difficile infection risk
Infectious/traveler's diarrhoea	Shortens duration and severity
IBS	Reduces bloating, flatulence, improves stool regularity
IBD	Adjunctive therapy in UC and pouchitis
Lactose intolerance	Aids digestion via bacterial lactase
H. pylori infection	Adjuvant to triple therapy — may reduce side effects, improve eradication

Generally safe, especially in healthy individuals. Use with caution in immunocompromised patients, critically ill/ICU patients, and those with central venous catheters (sepsis risk).

6 Irritable bowel syndrome (IBS) pharmacotherapy

IBS-C	Linaclotide (guanylate cyclase-C agonist → ↑cGMP → Cl^-/HCO_3^- secretion; ADR: diarrhoea), lubiprostone (↑intestinal fluid secretion, approved for women), plecanatide (similar to linaclotide), tenapanor (blocks Na^+ absorption), tegaserod (restricted use, women <65, no CV risk factors).
IBS-D	Rifaximin (antibiotic, ↓bacterial overgrowth), eluxadoline (mixed μ/κ agonist + δ antagonist, low oral bioavailability → limited CNS opioid effects, controlled substance), alosetron (severe cases only, women, rare serious ADRs), loperamide (OTC, short-term), bile acid binders (cholestyramine).
Pain & bloating	Antispasmodics (dicyclomine, hyoscyamine), peppermint oil, TCAs (amitriptyline — low dose, helps diarrhoea-predominant IBS), SSRIs (fluoxetine — when anxiety/depression coexist), neuropathic agents (pregabalin, gabapentin) for severe pain/bloating.

P Must-remember pearls

- Sulfasalazine is split by colonic bacteria into sulfapyridine (absorbed, causes systemic ADRs) and 5-ASA (acts locally in the lumen, anti-inflammatory).
- Thiopurines (azathioprine, 6-MP) have a delayed onset of 8–12 weeks; TPMT polymorphisms raise toxicity risk (myelosuppression).
- Methotrexate is more effective in Crohn's disease than ulcerative colitis; it inhibits dihydrofolate reductase.
- Anti-TNF agents (infliximab, adalimumab) risk TB and hepatitis B reactivation; vedolizumab has a safer systemic profile with much lower PML risk than natalizumab.
- Long-term 5-ASA therapy in extensive UC reduces colorectal cancer risk by about 75%.
- Diarrhoea is a common side effect of olsalazine specifically; Stevens-Johnson syndrome can occur with mesalamine/olsalazine/balsalazide.
- Use probiotics cautiously in immunocompromised, critically ill/ICU patients, and those with central venous catheters (sepsis risk).
- Eluxadoline (IBS-D) has low oral bioavailability, limiting central opioid effects; linaclotide (IBS-C) commonly causes diarrhoea as its main adverse effect.

Q Test yourself

1. A patient with ulcerative colitis asks how it differs from Crohn's disease. Which feature is characteristic of ulcerative colitis rather than Crohn's disease?

A. Continuous inflammation limited to the colon and rectum, with the rectum universally involved
B. Transmural injury that may lead to strictures or fistulae
C. Skip lesions with patchy involvement
D. Involvement of any part of the GI tract from mouth to anus
E. Insidious onset

2. A patient taking sulfasalazine for ulcerative colitis develops nausea, headache, and bone marrow suppression. Which component of the drug is responsible for these systemic side effects?

A. 5-aminosalicylic acid, which acts through the lumen
B. Sulfapyridine, which is absorbed systemically
C. The azo bond linking the two components
D. Colonic bacterial azoreductase itself
E. The enteric coating of the tablet

3. A patient with Crohn's disease is started on azathioprine. The lecture notes a delayed onset of action. Approximately how long does it take for thiopurines to become effective?

A. 24–48 hours **B.** 3–5 days **C.** 8–12 weeks **D.** 2–3 weeks **E.** 6–12 months

4. A patient with moderate-to-severe Crohn's disease is started on infliximab. What is this drug's mechanism of action?

A. It inhibits Janus kinases, blocking STAT signaling
B. It blocks α4β7 integrin, preventing lymphocyte trafficking
C. It binds the p40 subunit of IL-12/IL-23
D. It binds TNF-α, preventing its interaction with TNF receptors and inhibiting pro-inflammatory signaling
E. It inhibits dihydrofolate reductase

5. A woman under 65 with IBS-C has not responded to other treatments and has no cardiovascular risk factors. Which drug, reintroduced under specific restrictions due to past cardiovascular safety concerns, may be considered for her?

A. Rifaximin **B.** Loperamide **C.** Alosetron **D.** Eluxadoline **E.** Tegaserod

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — A — UC causes continuous mucosal/submucosal inflammation of the colon and rectum (rectum universally involved), while CD causes patchy transmural disease anywhere in the GI tract with skip lesions, strictures, and fistulae.

2 — B — Sulfasalazine is split by colonic bacteria into 5-ASA (acts locally/through the lumen) and sulfapyridine, which is absorbed and causes most systemic side effects (nausea, headache, bone marrow suppression, hypersensitivity reactions).

3 — C — Thiopurines (azathioprine, 6-MP) have a delayed onset of action of about 8–12 weeks, since they act by being converted to 6-thioguanine nucleotides that block lymphocyte DNA/RNA synthesis.

4 — D — Infliximab and adalimumab are anti-TNF agents that bind TNF-α, preventing it from interacting with its receptor and inhibiting pro-inflammatory signaling; ADRs include infusion reactions, infections (especially TB, hepatitis B reactivation), and rarely lymphoma.

5 — E — Tegaserod was reintroduced under restrictions for women under 65 with IBS-C who failed other treatments and have no cardiovascular risk factors, due to earlier cardiovascular safety concerns.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam